Exact minimum Kochen-Specker sets inside coordinate pools: certified optimality of the 31-, 33-, 18- and 29-ray sets in dimensions three, four and five

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Version 1.0.0, 10 November 2026. Research preprint, not peer reviewed.

> **What this paper does and does not do.** The smallest Kochen-Specker (KS) set in dimension three is an open problem: it has between 24 and 31 rays. This paper does **not** solve that problem and does **not** find a KS set smaller than any published one. It answers a restricted question exactly: *inside a given finite pool of rays, what is the smallest KS set?* For the pools listed in Table 2 the answer is the known record (31, 33, 18, 29 rays), and each answer comes with a certificate that can be re-checked without trusting the search programs: Lean 4 kernel proofs for two pools, replayable search trees, and unsatisfiability proofs checked by an independent proof checker for the others. Some of the smaller cases were reported before (Section 1.3); what is new is stated there, with the limits of the literature search that supports the word "new".
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> **How it was made.** The programs, proofs and text were produced by an AI system (Claude, Anthropic) at the request and under the direction of the author, who is responsible for the release (Section 9).

Abstract

A Kochen-Specker set is a finite set of rays that admits no assignment of 0 and 1 with exactly one 1 in every orthogonal basis and no two orthogonal rays both 1. We study the minimum size of such a set inside a prescribed finite pool of rays. A simple duality (every KS set must contain a basis missed by any set of pairwise non-orthogonal rays) turns the problem into a covering problem whose constraints are individually trivial to verify, and two necessary conditions for critical KS sets cut the candidate space sharply. We implement three independent exact methods and attach certificates to every result. Results: (1) among the rays of dimension three with integer coordinates of absolute value at most 2, 3 or 4, the minimum is 31, and for bounds 2 and 3 the minimum sets are exactly the six images of one Conway-Kochen-type set under the symmetries of the cube; (2) among the rays whose coordinates are 0 or N-th roots of unity, N = 6, 12, 18, 24, the minimum is 33, unique up to symmetry for N = 6 and 12, and for N = 2, 3, 4, 5, 7, 8, 9, 10 there is no KS set at all; (3) among the 40 rays of dimension four with coordinates 0, ±1 the minimum is 18, attained by exactly 16 sets forming one orbit; (4) among the 121 rays of dimension five with coordinates 0, ±1 the minimum is 29, so the 2005 record of Cabello, Estebaranz and García-Alcaine is optimal in that pool. Statements (1) for bound 2 and (3) are proved in Lean 4 and checked by its kernel, without Mathlib. Randomised searches in eleven further pools in dimensions three to eight reproduce every published record within seconds and find nothing smaller. The general minimal-KS problem is not settled by any of this.

1. Introduction

1.1 The Kochen-Specker theorem and its minimal sets

Kochen and Specker [1] and Bell [2] showed that quantum theory is incompatible with any model that assigns each projective measurement a pre-existing outcome independent of which other compatible measurements are made with it. The combinatorial core is a finite set of rays in a Hilbert space of dimension $d \geq 3$ that cannot be 0/1-colored: no assignment gives exactly one 1 to each orthogonal basis while never giving 1 to two orthogonal rays. Such a set is a *Kochen-Specker set*. The original set had 117 rays in dimension three; smaller sets followed, and the search for the smallest one in each dimension became a problem in its own right, partly because KS sets are the resource behind state-independent contextuality, pseudo-telepathy games and bipartite perfect quantum strategies [18, 20].

1.2 State of knowledge (meta-analysis)

Table 1 collects the smallest known KS set and the best proven lower bound in each dimension, as far as we could establish them from the primary sources on 9 November 2026. Counts follow the standard definition (Section 2): only the rays of the set are counted. Papers that use hypergraphs with all bases completed report larger numbers for the same sets in dimension three (the 31-ray set becomes "51-37").

Table 1. Smallest known KS sets and proven lower bounds.

d	smallest known	source	proven lower bound	source
3	31 (real)	Conway and Kochen, in [3]	24 (real and complex)	[10]; independently, real case, [11]
3	33 (complex, 14 bases)	[20, 21]		
4	18	[5]	18	[7, 12]
5	29	[6]	18	[12]
6	21 (complex), 31 (real)	[13], [6]	18	[12]
7	32 (complex), 34 (real)	[15], [6]	18	[12]
8	34	[14]	18	[12]

Three features of the table matter here. First, only dimension four is settled. Second, in dimension three the gap 24 to 31 has resisted large computations: the lower bound 24 came from exhaustive enumeration of candidate orthogonality graphs up to 23 vertices, with the unembeddability step done by an SMT solver [10, 11], and the authors of [11] estimate the next order at more than a hundred CPU years. Third, in dimensions five and above the only proven bound is the dimension-independent 18 of [12], so the gaps are wide.

Restricted results exist in dimension three. Trandafir and Cabello [18] proved that the Conway-Kochen set is rigid and that no KS set with 30 or fewer rays arises from the 13-ray Yu-Oh set [26] by their completion procedure, and conjectured that 31 is the minimum. Li, Bright, Trandafir, Cabello and Ganesh [19] enumerated, with proof certificates, all KS sets with up to 33 rays containing the 25-ray completion of the Yu-Oh set. Pavičić and Megill [16, 17] generate very large families of critical sets from all vectors with components in a small set; their smallest sets are found by sampling and are upper bounds.

1.3 The pool problem, prior work on it, and what is new here

We fix a finite *pool* of rays and ask for the exact minimum size of a KS set inside it. This is weaker than the general problem (a pool result says nothing about rays outside the pool) but it is exactly solvable, it gives matching upper and lower bounds, and for natural pools it tests the conjectured optimality of the records where they live.

The closest prior work known to us is Kernaghan's survey of coordinate alphabets in dimension three [22], a preprint that states its computations and text were generated by a language model. It reports, by SAT with unsatisfiable-subset extraction, the minimum 31 for the 49 rays with coordinates in {0, ±1, ±2} and the six minimal 31-ray sets there, the minimum 33 for the 57-ray pool of sixth roots of unity and for some other two-symbol alphabets, and lists 31 for coordinates up to 3; it gives no proof certificates and treats neither full N-th-root alphabets for N > 6 nor any dimension above three. An online forum note of October 2026 reports 31 for integer coordinates up to 4 without certificates (see Section 7 for what we could and could not confirm about it). We therefore claim the following as new, to our knowledge and after the bounded search described in Section 7:

- certificates for all of these statements, including Lean 4 kernel proofs for the 49-ray pool and the 40-ray pool in dimension four (Section 5), and checked unsatisfiability proofs elsewhere;
- uniqueness up to symmetry of the minimum sets for integer coordinates up to 3, and for the sixth- and twelfth-root pools;
- the root-of-unity pools N = 12, 18, 24 (minimum 33);
- the first exact minimum in a natural pool beyond dimension four: 29 in $\{0, \pm 1\}^5$;
- the method in this form: coloring cuts combined with necessary conditions for critical sets, with three independent engines and certificates designed to be checked by small programs.

We do not claim the values 31, 33, 18, 29 themselves, which are the published records [3, 20, 5, 6].

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2. Definitions

A *pool* is a finite set V of rays (one-dimensional subspaces) of a d-dimensional real or complex inner-product space, d ≥ 3. Two rays are *orthogonal* if their inner product vanishes. A *basis* is a set of d mutually orthogonal rays of V.

A subset S of V is *colorable* if there is $f : S \to \{0, 1\}$ such that

1. $f(u) + f(v) \le 1$ whenever $u, v \in S$ are orthogonal, and

2. every basis contained in S has a ray with f = 1.

Otherwise S is a *Kochen-Specker set*. By rule 1 a basis inside S then has exactly one ray of value 1. Only the rays of S count; orthogonal pairs constrain f whether or not they extend to a basis inside S. Subsets of colorable sets are colorable. A KS set is *critical* if all its proper subsets are colorable; every KS set contains a critical one, so the minimum is attained by a critical set.

The pools used here are: B(d, H), the rays spanned by non-zero integer vectors of Z^d with all coordinates of absolute value at most H; and U(d, N), the rays spanned by non-zero vectors of C^d with entries in $\{0\} \cup \mu_N$, where μ_N is the group of N-th roots of unity. Orthogonality is decided exactly: by integer dot products in B(d, H), and in U(d, N) by reducing the inner product, an integer combination of N-th roots of unity, modulo the N-th cyclotomic polynomial.

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3. Method

Full proofs of the statements in this section are in the repository file `paper/PROOFS.md`; they are short and are repeated here in outline.

3.1 Coloring cuts

Call $I \subseteq V$ *independent* if no two of its rays are orthogonal, and let Miss(I) be the set of bases of V containing no ray of I.

Lemma 1. $S \subseteq V$ is a KS set if and only if, for every independent $I \subseteq V$, S contains a basis of Miss(I).

If S contains no basis of Miss(I), then giving 1 to the rays of I inside S colors S. Conversely the rays of value 1 of a coloring form an independent set I with no basis of Miss(I) inside S.

So every independent set I yields a *cut*, the clause "S contains at least one basis of Miss(I)", valid for every KS set and certified by I alone: one checks that I is independent and recomputes Miss(I). Symmetries of the pool map cuts to cuts. A consequence: if $C \subseteq V$ is colorable, every KS set meets the complement of C.

3.2 Necessary conditions for critical sets

Lemma 3. If T is a critical KS set and $v \in T$, then (C1) v lies in a basis contained in T, and (C2) v is orthogonal to at least d rays of T.

For (C1), a coloring of T without v extends by f(v) = 0 if no basis through v lies in T. For (C2), if v had at most $d - 1$ partners they would be the rest of the unique basis b through v, and f(v) = 1 if the rest of b is all 0, f(v) = 0 otherwise, extends the coloring.

Applied to the pool itself, the lemma allows the repeated deletion of rays that lie in no basis or have fewer than d partners; what remains is the *core*. In the pools of Pavičić and Megill the core has the size of their "master" sets,

which gives an external check of our pool construction: 105 rays and 136 bases for B(5, 1), 332 and 1408 for B(6, 1), 805 and 9936 for B(7, 1), 834 and 1609 for U(6, 3), as in [16, 17].

3.3 Counterexample-guided search

Take Boolean variables x_v (ray chosen) and y_b (basis chosen) with y_b true exactly when all rays of b are chosen. For a bound k the *candidate problem* asks for an assignment with at most k rays chosen that satisfies (C1), (C2) and all cuts collected so far. A solution S is tested: if it is not colorable, a KS set with at most k rays has been found. Otherwise a coloring of S is extended to an independent set of the whole pool that misses as few bases as possible; its cut is violated by S and is added, together with its images under the symmetries of the pool. If the candidate problem becomes infeasible, the pool contains no KS set with at most k rays, by Lemmas 1 and 3.

3.4 Search trees

For small pools we also use a plain exhaustive search whose trace is a certificate. A node holds an independent set I and one child for each basis of Miss(I) that could still be added within the budget k without completing a basis already excluded on the way; a child's subtree treats the sets that contain that basis and none of the earlier siblings' bases. A node without children is closed because I colors every remaining admissible set. Lemma 5 of `paper/PROOFS.md` states the checking rule and proves by induction that an accepted tree shows every set of at most k rays to be colorable. The rule uses neither (C1) nor (C2). The dual certificate, that one given set is not colorable, is a case tree on which ray of a basis has value 1.

3.5 Symmetry and cases

In B(d, H), signed coordinate permutations are symmetries. Every set can be brought to a normal form in which the coordinates are sorted by the number of chosen rays with a non-zero entry there and, for j = 1, …, d − 1, the ray $e_0 - e_j$ is chosen only if $e_0 + e_j$ is (Lemma 6). For B(5, 1) we also use:

Lemma 7. Every basis of B(5, 1) contains a coordinate axis, and every KS set in B(5, 1) contains at least three axes.

The second part holds because the pool with any three axes removed is colorable (an explicit coloring is in `results/lemma7_d5_axes.json`). A critical KS set therefore has, up to a coordinate permutation, exactly the axes $\{e_0, e_1, e_2\}$, or $\{e_0, e_1, e_2, e_3\}$, or all five, and the three cases are searched separately. The last case is further split into 81 sub-cases by the states of the four pairs $(e_0 + e_j, e_0 - e_j)$.

3.6 Three engines and what has to be trusted

- **Engine A** solves the candidate problem with the CP-SAT solver of OR-Tools [29].
- **Engine B** re-encodes the final candidate problem from scratch as a propositional formula in conjunctive normal form, re-verifying each cut from its independent set and using two-sided sequential counters for the counting constraints, and solves it with the CDCL solver Glucose [28] through PySAT [27], which emits a DRUP proof [30]. The proof is checked by `verify/drup_check.c`, a forward reverse-unit-propagation checker of about 200 lines written for this project and tested on correct proofs, corrupted proofs and satisfiable formulas.

- **Engine C** is the exhaustive tree search of Section 3.4. Its certificates are replayed by an independent JavaScript program (`verify/verify_cert.mjs` , written from the specification alone, which also recomputes orthogonality and all bases from the integer vectors) and, for two pools, by a checker whose soundness is proved in Lean 4.

For a result marked "proof-checked", what must be trusted is: the pool construction, the encoder, the cut verification, the proof checker, and Lemmas 1, 3, 6, 7. The solvers are not trusted. For a result marked "Lean", what must be trusted is the Lean kernel and the reading of the definitions in `lean/KS/Basic.lean` .

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4. Results

Table 2. Exact minimum size of a KS set inside each pool.

Pool	rays (core)	bases	minimum	minimum sets	evidence
B(3, 2): integer coordinates ≤ 2	49 (49)	26	31	6, one orbit	Lean; tree (255 nodes); engines A, B; proof-checked
B(3, 3)	145 (97)	50	31	6, one orbit	tree (11 247 nodes); engines A, B; proof-checked
B(3, 4)	289 (205)	110	31	see Section 4.1	engines A, B
U(3, 6): sixth roots of unity	57 (57)	22	33	3, one orbit	exhaustive tree search; engine A
U(3, 12)	183 (183)	67	33	12, one orbit	engines A, B; proof-checked
U(3, 18)	381 (381)	136	33	not enumerated	engine A
U(3, 24)	651 (651)	229	33	not enumerated	engine A
B(4, 1): coordinates 0, ±1	40 (40)	32	18	16, one orbit	Lean; tree (509 nodes); engines A, B; proof-checked
B(5, 1)	121 (105)	136	29	not enumerated	engine B with case analysis, Section 4.4

The file `results/RESULTS.md` in the repository is the register behind this table and records the status of every certificate, including those still running when this version was written.

4.1 Dimension three, integer coordinates

Theorem 1. Let H ∈ {2, 3, 4}. Every KS set contained in B(3, H) has at least 31 rays, and 31 is attained. For H ∈ {2, 3} there are exactly six KS sets with 31 rays; they have coordinates of absolute value at most 2 and are the images of

one of them under signed coordinate permutations.

The 31-ray set has 71 orthogonal pairs and 17 bases, the parameters of the Conway-Kochen set; the count of six agrees with [18] and [22]. For H = 2 the statement "minimum 31" is the Lean theorem `KS.Box3H2.minimum_ks_size`. Enlarging the pool from 49 to 145 rays adds 48 rays to the core and no new minimum set. The best coloring of the whole pool misses 1 of 26 bases for H = 2, 2 of 50 for H = 3, 7 of 110 for H = 4 and 14 of 266 for H = 5, a measure of how fast the pools fill with obstructions. For H = 5 (577 rays, core 421) engine A did not finish in 78 minutes and we make no claim.

4.2 Dimension three, roots of unity

Theorem 2. For N ∈ {6, 12, 18, 24} every KS set contained in U(3, N) has at least 33 rays, and 33 is attained. In U(3, 6) there are exactly 3 KS sets with 33 rays and in U(3, 12) exactly 12; in both cases they form a single orbit under coordinate permutations, multiplication of coordinates by roots of unity and complex conjugation. For N ∈ {2, 3, 4, 5, 7, 8, 9, 10} the pool U(3, N) contains no KS set.

The 33-ray sets have 78 orthogonal pairs and 14 bases, the parameters of the complex set of [20, 21]. The value 33 for the 57-ray pool U(3, 6) was reported in [22]. Among the tested N a KS set exists exactly when 6 divides N; we have not proved this for all N.

4.3 Dimension four

Theorem 3. Every KS set contained in B(4, 1) has at least 18 rays; there are exactly 16 KS sets with 18 rays, forming one orbit under signed coordinate permutations.

The bound 18 holds for all KS sets in every dimension [7, 12]; the theorem is included because its Lean proof (`KS.Box4H1.minimum_ks_size`) is, to our knowledge, the first kernel-checked proof of a minimality statement for KS sets, and because the count 16 matches the known number of 18-ray sets inside the 24 rays of Peres.

4.4 Dimension five

Theorem 4. Every KS set contained in B(5, 1) has at least 29 rays, and 29 is attained.

Proof structure. By Lemma 7 and the normal form there are three cases for a critical KS set with at most 28 rays. Table 3 lists how each was closed. The 29-ray set found by search has 16 bases and 151 orthogonal pairs and contains exactly three axes; it consists of an 18-ray KS set in a coordinate hyperplane and 11 further rays, the structure of the recursive construction of [6].

Table 3. The three cases of Theorem 4 (bound k = 28).

Case	engine A (CP-SAT)	engine B (Glucose)	proof check
exactly three axes	infeasible (two runs)	unsatisfiable	VERIFIED, 94 680 lemmas
exactly four axes	did not finish	unsatisfiable	see `results/RESULTS.md`

Case	engine A (CP-SAT)	engine B (Glucose)	proof check
all five axes	did not finish	81 sub-cases, all unsatisfiable	see `results/RESULTS.md`

Lemma 7 has a corollary of independent interest: a KS set built from vectors with entries 0, ±1 in dimension five always contains at least three coordinate axes.

4.5 Searches for smaller sets

A randomised walk on critical KS sets (remove a few rays, then repair to a critical KS set that keeps as much of the old set as possible) was run for 10 to 15 minutes in each of the pools of Table 4. In every pool it reached the published record, usually within seconds, and nothing smaller.

Table 4. Randomised searches (no optimality claim).

Pool	rays in core	smallest KS set found	published record
B(3, 2)	49	31	31 [3]
U(3, 12), U(3, 18)	183, 381	33	33 [20, 21]
B(4, 1), U(4, 6)	40, 400	18	18 [5]
B(5, 1), B(5, 2)	105, 1185	29	29 [6]
U(5, 4), U(5, 6)	525, 1505	29	29 [6]
B(6, 1)	332	31	31 real [6]
U(6, 3)	834	21	21 [13]
B(7, 1)	805	34	34 real [6]
U(7, 3)	2968	32	32 [15]
E8 root rays (d = 8)	120	34	34 [14]

This is evidence, not proof, that the records are optimal in these pools. In particular we did not decide whether 21 is the minimum in U(6, 3) or 34 in the E8 pool; both exact searches were stopped unfinished.

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5. Formal verification in Lean 4

The Lean development (`lean/` , Lean 4.35.0, no Mathlib) has three layers.

1. `KS/Basic.lean` defines pools as tables of bit masks, colorings, and the two certificate checkers `check` (Section 3.4) and `refute` , and proves their soundness: `KS.small_sets_colorable` (an accepted tree certificate implies that every set of at most k rays is colorable) and `KS.not_colorable_of_refute` (an accepted refutation implies

that the given set has no coloring). While proving them, the first versions of both checkers were found to accept malformed certificates; the definitions were corrected and the counterexamples are now rejected.

2. Each data module lists the integer vectors, the orthogonality table and the bases, and proves by kernel evaluation that the table is "dot product zero" on the vectors and that the bases are exactly the sets of d mutually orthogonal rays.
3. The certificates are evaluated by the kernel (`decide +kernel`), giving

```
theorem KS.Box3H2.minimum_ks_size :
    (∃ T, T < 2 ^ 49 ∧ card 49 T = 31 ∧ ¬ Colorable pool T) ∧
    (∀ T, T < 2 ^ 49 → card 49 T ≤ 30 → Colorable pool T)
```

and the analogous `KS.Box4H1.minimum_ks_size` with 40, 18 and 17. Sets of rays are natural numbers read as bit masks. The axioms used are `propext` , `Classical.choice` and `Quot.sound` ; there is no `sorry` and no `native_decide` . Checking either data module takes under a minute.

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6. An operational check on the Tsotchke stack

The 31-ray set was also exercised on four independent pieces of the Tsotchke software stack, all classical programs: the Moonlab state-vector simulator (WebAssembly build), the Fubini-Study kernel of the quantum geometric tensor library, the integer mixers of quantum_rng as a reproducible pseudo-random stream, and an exact-integer program in the Eshkol language. A qutrit is embedded in two simulated qubits. The 71 orthogonal pairs and 17 bases are recovered from simulated overlaps and from Fubini-Study distances equal to $\pi/2$; for 2000 pseudo-random states the 17 basis sums of outcome probabilities equal 1 to within $2 \cdot 10^{-15}$, so the sum over bases is 17 for every state, whereas an exhaustive search shows that an assignment obeying rule 1 gives a 1 to at most 16 of the 17 bases. This restates the state-independent gap of the set in operational terms; it is a simulation, not an experiment, and no quantum randomness is involved.

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7. Limits, prior art, open questions

- **Scope.** A pool result excludes smaller KS sets only inside the pool. The general problem in dimension three (24 to 31) and in dimensions five and above (18 to the records of Table 1) is open.
- **Prior art and the word "new".** Literature was searched on 9 and 10 November 2026 on arXiv, Zenodo, GitHub and the open web, by topic and by the distinctive numbers. Reference [22] anticipates the values for B(3, 2), U(3, 6) and, without certification wording, B(3, 3). The forum note mentioned in Section 1.3 could not be verified beyond its own text. Preprints in this area appear weekly; an overlap we did not find is possible.
- **What is not certified to the same level.** Rows of Table 2 that list only engine A rest on one solver without an independent proof check at the time of writing; `results/RESULTS.md` records later checks. The randomised searches prove nothing about optimality.

- **Open questions.** Is 31 the minimum in B(3, H) for all H? Does U(3, N) contain a KS set exactly when 6 divides N, always with minimum 33? Is 21 the minimum in U(6, 3)? Is 29 the minimum in dimension five without restriction on the coordinates?

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8. Reproducibility

Repository: https://github.com/0thernes/kochen-specker-extremal-lab (site: https://0thernes.github.io/kochen-specker-extremal-lab/). `python scripts/run_all.py` runs the unit tests, regenerates the two small tree certificates, replays them with the JavaScript verifier, regenerates and checks the unsatisfiability proofs for the two smallest pools, and re-verifies Lemma 7; `--lean` adds the Lean build and axiom audit, `--stack` the demonstration of Section 6. Commands and logs for every row of Table 2 are indexed in `docs/INDEX.md`. Software: Python 3.9, PySAT 1.8, OR-Tools 9, Node 25, gcc 16, Lean 4.35.0.

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9. Disclosure, funding, licence

The research, programs, Lean proofs and text were produced by Claude (Anthropic) working as an agent at the request and under the direction of the author, who takes responsibility for the release. An AI system is not an author. The work was funded by the author through 0thernes LLC; there was no external funding and there are no competing interests. The repository is released under the verification licence in its `LICENSE` file; third-party components keep their own licences.

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